

Bernard Balko

Mechatronics & Robotics Engineer

New York, NY • (860) 515-6415 • bernard.balko1@gmail.com • [bernardbalko.github.io](https://github.com/bernardbalko) • linkedin.com/in/bernard-balko

EDUCATION

New York University, Tandon School of Engineering

2026 – Present

M.S. Mechatronics & Robotics (Currently Enrolled)

Coursework: Foundations of Robotics, Mechatronics, Embedded Systems, Control Systems, Sensor Fusion

University of Hartford

May 2022

B.S. Mechanical Engineering | Minor in Mathematics

Dean's List: Spring 2019 – Spring 2021 • Senior Capstone published at ASME IMECE 2022

TECHNICAL SKILLS

Robotics & Control: Forward/Inverse Kinematics, Jacobian, DH Parameters, PD/Proportional Control, Trajectory Tracking, Sensor Fusion, MuJoCo

Embedded Systems: STM32, Arduino, Embedded C, I2C/ADC, IMU (MPU6050), Haptic/LED Feedback, Signal Filtering

Programming: Python, C++, C, MATLAB, OpenCV, PySerial, scipy

3D Design & CAD: Siemens NX, SolidWorks, OnShape, AutoCAD, GD&T, DFA, 3D Printing

Simulation & Analysis: MuJoCo, COMSOL, ANSYS Workbench, FEA, Root Cause Analysis

Perception & Vision: OpenCV, Haar Cascade Classification, Real-Time Video Processing, LiDAR, ICP, MSAC/RANSAC, Point Cloud Processing, Voxelization

PROJECTS

Real-Time Face-Tracking Turret (Computer Vision + Embedded Control)

2026

- Engineered a full-stack mechatronic system achieving real-time face tracking across a physical pan/tilt turret by integrating a Python/OpenCV vision pipeline with an STM32 Nucleo over a 115,200-baud UART serial link driving dual PWM-controlled servos (1000–2000 μ s)
- Eliminated servo jitter and false-positive target lock by implementing an asymmetric deadzone filter (35 px pan / 55 px tilt) and a spatial proximity tracking algorithm that validates detections against a 180-pixel Euclidean history radius, suppressing environmental noise that caused violent oscillation
- Stabilized a positive feedback loop causing target loss by diagnosing an inverted tilt direction sign and calibrating the proportional gain to 0.04, then implemented an automated state machine that recovers to a default center position (1500 μ s) after 10 consecutive frames of lost lock

Smart Posture Trainer (Wearable Mechatronic System) |

2026

- Designed a real-time wearable posture monitor using an MPU6050 IMU and flex sensor on a chest strap controlled by an STM32H503RB microcontroller
- Implemented dual-sensor fusion via I2C/ADC with low-pass filtering and calibration-based baseline to eliminate false positives from natural movement
- Triggered haptic (vibration motor) and visual (LED) feedback only when poor posture persisted beyond 1 second — demonstrating embedded control logic and interrupt-driven design

UR10e 6-DOF Robot Kinematics — MuJoCo Simulation |

2026

- Implemented full forward and inverse kinematics (position & velocity) from scratch for the UR10e manipulator inside a MuJoCo physics simulation
- Derived homogeneous transformation matrices from DH parameters; solved nonlinear IK using scipy optimization
- Computed the Jacobian for velocity-level kinematics and implemented a PD torque controller achieving stable joint-angle tracking

Aerial Vehicle Rapid 3D Map Generation — ASME Published |

2022

- Designed a LiDAR scanning turret on a pan-tilt mechanism with IMU feedback for autonomous safe-landing zone detection in UAVs

- Applied ICP point cloud stitching and MSAC plane segmentation to classify flat, wall, and non-flat terrain surfaces
- Benchmarked 1×1, 5×5, and 10×10 voxelization grids for optimal speed/accuracy trade-off; results published at ASME IMECE 2022

WORK EXPERIENCE

Mechanical Design Engineer | *Quest Global — Pratt & Whitney (Remote)* June 2022 – Dec 2025

- Led a team of 3 engineers on F135/F119 engine inspection documentation, improving process efficiency by 30% and doubling project scope
- Conducted GD&T analysis on F135 STOVL 3BSM engine components, identifying risks and developing mitigation strategies to improve product reliability
- Performed root cause analysis on nonconforming PW4000-100, PW4000-94, PW2000, and F117 parts; implemented corrective actions and cross-functional process improvements
- Developed custom Excel tools for bearing shear stress and tolerance stack-up calculations, increasing precision in part assessments
- Used Siemens NX for 3D modeling and structural analysis; collaborated with structural, manufacturing, and aero engineering teams

PUBLICATIONS

Tatoglu A., Yin C.C., Cervello B., Corrado A., Balko B., Sohn K. "Aerial Vehicle Rapid 3D Map Generation for Safe Landing Area Detection." ASME IMECE 2022, Paper no. IMECE2022-95849. Columbus, Ohio, Oct–Nov 2022.

LANGUAGES

English (Native) • Korean (Proficient)